

Contact

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www.linkedin.com/in/valiantsin-kasabrukhou (LinkedIn)

Top Skills

Wiring Diagrams

Soldering

Knowledge Engineering

Languages

Russian (Native or Bilingual)

French (Professional Working)

Polish (Limited Working)

English (Native or Bilingual)

Honors-Awards

American Savings Foundation
Scholar

AP Scholar with Distinction

AP Scholar with Distinction

Connecticut Association of Boards
of Education Student Leadership
Award

2024 Radcliff Plainville High School
Fund

Valiantsin Kasabrukhou

Duke CS+AI / Math '28 | Corporate SWE @ Travelers
Plainville, Connecticut, United States

Summary

I'm a developer who builds cool things and has a deep passion for technology, research, science, education, and energy (sometimes a little bit of history too). I currently attend Duke University, where I am majoring in Computer Science and Math, along with a minor in Earth and Climate Sciences.

My current personal interests are split between Nuclear Energy, Quantum Computing, and AI in education. If you're up to discuss any of those areas of knowledge, please reach out, I'm always open to a good conversation!

Experience

Duke University Pratt School of Engineering

Undergraduate Researcher

December 2025 - Present (8 months)

Durham, North Carolina

Helping build incredible experiences at the I3T with Dr. Maria Gorlatova

Duke Applied Machine Learning Group

Developer

September 2025 - Present (11 months)

Durham, North Carolina, United States

Working as part of the DAML executive team to remodel the online member experience of 50+ members.

- Custom building a brand new mobile and web experience for members and program partners
- Actively prototyping new user experience in Figma, to then later implement on the MERN stack
- Working on creating a new mobile event experience portal

Travelers

EDP Software Engineer

June 2025 - Present (1 year 2 months)

Hartford, Connecticut, United States

Supported in-house tools for 34,000+ employees across the enterprise.

- Prototyped AI-driven tools & automation pipelines on using AWS's TextRact & Models on Bedrock, boosting underwriting efficiency for 11,000+ employees.
- Advised development of educational curriculum for 100+ Engineering Development Program interns.
- Built B2B AWS infrastructure (C2 Instances, S3 Containers, Lambdas, API Gateways) to power internal tools & applications.
- Pitched product ideas to multiple department CIOs, VPs, & Head Architects.
- Optimized API queries and response guidelines, cutting response times by up to 30% while strengthening data security.
- Wrote Terraform declaration and YAML guidelines for APIs and user applications. Subsequently created JUnit tests for all deliverables.
- Established Snyk & 42Crunch integrations for CI/CD pipeline security.
- Designed UX/UI prototypes in Figma & leveraged OpenAI/Anthropic APIs to prototype an enterprise project management platform.
- Updated front-end user experiences via React, assisted with code review, and frequently pushed to prod.

Nicholas Institute for Energy, Environment & Sustainability

1 year 11 months

Researcher

January 2025 - Present (1 year 7 months)

Durham, North Carolina, United States

Co-authoring a senior honors thesis with Prof. Cinq-Mars on post-USSR immigrant migration to Silicon Valley and the influence of Soviet cultural values on U.S. technological development.

- Analyzing 30+ years of Silicon Valley history to quantify the impact of USSR immigrant contributions on American innovation.
- Conducting original Russian-language research through primary sources, archival texts, and interviews with key figures.
- Generating new primary resources by interviewing Russian-speaking immigrant founders, engineers, and community leaders.
- Developing original narratives and frameworks connecting cultural heritage with U.S. technological and economic growth.

Research Assistant

September 2024 - January 2025 (5 months)

Durham, North Carolina, United States

I worked as a research assistant under Tom Cinq-Mars at Duke University.

- Translated, reviewed, and analyzed Russian literature in the fields of energy, economics, biology, physics, history, and social studies.
- Assisted Professor Cinq-Mars in synthesizing the research as well as deriving the most useful resources for his research.
- Created citations and a bibliography for the project in Zotero
- Collaborated with the “Reexamining Nuclear Power in the Carolinas and Beyond” Bass Connections Research Team at Duke
- Conducted research on energy markets in the USSR, specifically in regards to the Druzhba pipeline.

Duke Robotics Club

Electrical Engineering Team Member

September 2024 - May 2025 (9 months)

Durham, North Carolina, United States

Contributed to electrical systems design and prototyping for autonomous underwater robotics.

- Accelerated prototyping and testing by replicating core robot circuitry onto dedicated prototyping boards alongside a 20–30 member electrical team.
- Enabled cross-team development of a mini-sub robot through collaboration with hardware and software teams on integrated system design.
- Advanced underwater communication research by contributing to the design of a new acoustic signaling system.

Acme Monaco

Database and AI Automation Software Engineer

April 2024 - January 2025 (10 months)

New Britain, Connecticut, United States

Delivered automation, database, and AI solutions supporting 300+ employees across CT, ME, and Singapore facilities.

- Reduced query times ~20% by restructuring company-wide MySQL infrastructure.
- Cut shop-floor processing time and improved FDA compliance by deploying custom Laserfiche automation for product spec forms.
- Automated inventory for 10,000+ items by integrating the Remstar auto-shelving system with Epicor ERP and MySQL pipelines.

- Reduced financial model turnaround from hours to minutes with Python + Pandas automation integrated into Excel workflows.
- Increased ERP efficiency and reduced manual entry through Epicor API automations for cross-system data sync.
- Advanced QA throughput by prototyping AI-powered guidewire inspection with PyTorch and TensorFlow, reducing reliance on Keyence machines.
- Accelerated production R&D with Fusion360 models for manufacturing tools, delivering rapid prototypes to the shop floor.
- Explored scalable enterprise apps via MERN stack prototypes and Python automation proofs of concept.

International Economic Alliance (IEA)

Web Application Developer

June 2024 - October 2024 (5 months)

Boston, Massachusetts, United States

Contributed to digital infrastructure and data-driven initiatives supporting global economic policy discussions.

- Improved event visibility for G20 representatives by optimizing Salesforce and WordPress systems, driving SEO campaigns for high-profile international conferences.
- Worked with G20 representatives for a curated event experience.
- Prototyped mobile applications leveraging Google Gemini, NFC-based profile exchange, and social networking features for country delegates.
- Enabled economic research integration by loading and processing international datasets into virtual notebooks, supporting analysis and product development with the Kellis Lab at MIT.

Plainville High School

Vex VRC Program Founder and Head Team Engineer

October 2022 - June 2024 (1 year 9 months)

Plainville, Connecticut, United States

- Worked closely with the Superintendent, Board of Education, and PHS Administrative Team to create a robotics competition program at Plainville High School.
- Assisted with raising funding via sponsorships and school funding, with a total value of upwards of \$10,000
- Won a Judges Award and mobilized 2-3 teams into numerous competitions in both CT & MA
- Established a digital networking portal for future program fluidity

Town of Plainville, CT

Student Representative

January 2022 - June 2024 (2 years 6 months)

Plainville, Connecticut, United States

- Selected by the Board of Education as the school system liaison/ representative for all 5 public schools in Plainville.
- Responsible for presenting at every board meeting and reporting on school issues and events.
- Advised Board of Education members directly on issues within school programs and on the performance of school administrators.
- Provided advisory regarding potential staff optimizations as well as student policy corrections.

theCoderSchool

Instructor

June 2023 - August 2023 (3 months)

Farmington, Connecticut, United States

- Instructed programming classes of up to 20 students at a time, taught a total cohort of over 150 students. Ages ranged from 7 to 13
- Organized custom project-based curriculum for week-long classes on relevant CS & Engineering Topics.
- Walked students through lectures, debugging sessions, organized group projects, and directed class-led initiatives, and provided an introduction to complex topics in CS and Math.

Town of Plainville, CT

Intern Information Technology Department

July 2021 - August 2021 (2 months)

Plainville, Connecticut, United States

- Responsible for the repair, maintenance, and upgrades of town-wide IT infrastructure.
- Administrative responsibilities with industry-grade software.
- Responsible for assisting with data management for all community schools, police departments, and town library IT infrastructure.

Education

Duke University

Bachelor of Science - BS, Computer Science · (March 2024 - May 2028)

University of Connecticut

World History, Engineering · (September 2021 - June 2024)

Plainville High School

High School Diploma, STEM Inclined · (August 2020 - June 2024)

Central Connecticut State University

Computer Science · (August 2022 - December 2023)